

OIS-005 — Schema Implementation

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-004
 - OAS-011 — Schema Architecture
 - OCS-000 through OCS-032
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Purpose

This specification defines the implementation responsibilities for Constitutional Schemas.

The Architecture defines the concept of a living Schema.

This specification defines how executable systems represent, validate, evolve, and exchange Schemas while preserving constitutional meaning.

Implementation SHALL execute Schemas.

Implementation SHALL never allow Schemas to become implementation artifacts.

A Schema SHALL remain a constitutional description of a Place.

1. Implementation Objective

The objective of Schema Implementation is to ensure that every Place possesses an executable representation of its constitutional organization while remaining independent of implementation technology.

Implementation SHALL preserve:

Identity.

Meaning.

Evolution.

Compatibility.

Validation.

Historical continuity.

2. Executable Representation

Every Schema SHALL possess an executable representation.

The representation MAY be implemented using:

Structured documents.

Semantic graphs.

Object models.

Relational mappings.

Type systems.

Domain models.

The representation SHALL preserve constitutional semantics.

3. Schema Registry

Every implementation SHALL maintain a Schema Registry.

The Registry SHALL preserve:

Canonical Schema identity.

Version lineage.

Compatibility relationships.

Migration history.

Applicable constitutional specifications.

The Registry SHALL remain reconstructable.

4. Schema Validation

Implementation SHALL validate constitutional consistency.

Validation SHALL evaluate:

Identity integrity.

Relationship integrity.

Architectural compatibility.

Constitutional compatibility.

Historical continuity.

Validation SHALL NOT redefine constitutional meaning.

5. Schema Evolution

Schema evolution SHALL preserve:

Canonical identity.

Historical lineage.

Migration compatibility.

Relationship continuity.

Semantic equivalence where applicable.

Every evolution SHALL become constitutional history.

6. Schema Compatibility

Implementations SHALL distinguish between:

Compatible evolution.

Breaking evolution.

Experimental evolution.

Deprecated evolution.

Compatibility SHALL remain explicit.

7. Schema Migration

Migration SHALL preserve:

Meaning.

Relationships.

Historical continuity.

Identity.

Architectural intent.

Migration SHALL remain reversible whenever constitutionally possible.

8. Schema Federation

Federated Places SHALL exchange Schemas while preserving:

Local sovereignty.

Canonical identity.

Architectural compatibility.

Independent evolution.

Schema federation SHALL remain version-aware.

9. Schema Verification

Verification SHALL demonstrate:

Semantic preservation.

Validation correctness.

Migration correctness.

Compatibility.

Equivalent constitutional organization.

Implementation representations MAY differ.

Schemas SHALL remain constitutionally equivalent.

10. Schema Pattern

Implementation SHALL treat Schemas as executable constitutional contracts rather than implementation-specific metadata.

Schemas SHALL govern implementation.

Implementation SHALL not govern Schemas.

11. Schema Implementation Test

Schema Implementation satisfies this specification when independent implementations can exchange, validate, evolve, migrate, and reconstruct constitutional Schemas while preserving equivalent constitutional organization despite differences in programming languages, storage technologies, and execution environments.

Conformance

An implementation conforms to this specification when Schemas remain executable, versioned, historically continuous, semantically equivalent, and constitutionally authoritative independently of implementation technology.

Implementation Principle

Traditional systems define software first and schemas second.

Organodynamics defines constitutional meaning first,

architectural organization second,

and executable Schemas third.

Implementation therefore exists not to invent structure,

but to faithfully execute a living constitutional description of every Place capable of continuous evolution.

End of Specification OIS-005